

Heron V6 timing and motion circuit

Document role: source-of-truth explanation of the V6 electrical design Revision: V6 logical architecture Updated: 2026-08-25 Applies to: 2 VLP-16s, 1 Xsens MTi, 4 FLIR Forge cameras, 1 BTS7960/IBT-2 motor driver, 1 A/B encoder, and 1 normally-closed limit switch

Update this file and timing_circuit.pdf whenever the V6 schematic, connector pinout, Teensy pin assignment, power architecture, or firmware safety contract changes.

Release status

This document defines the target V6 circuit. It is ready for schematic review, but it is not yet a fabrication release. Fabrication requires the older isolated/USB-only KiCad design to be replaced, whole-hierarchy ERC = 0, a fully routed board with DRC = 0 and zero unrouted connections, verified footprints, and completion of the bench gates below.

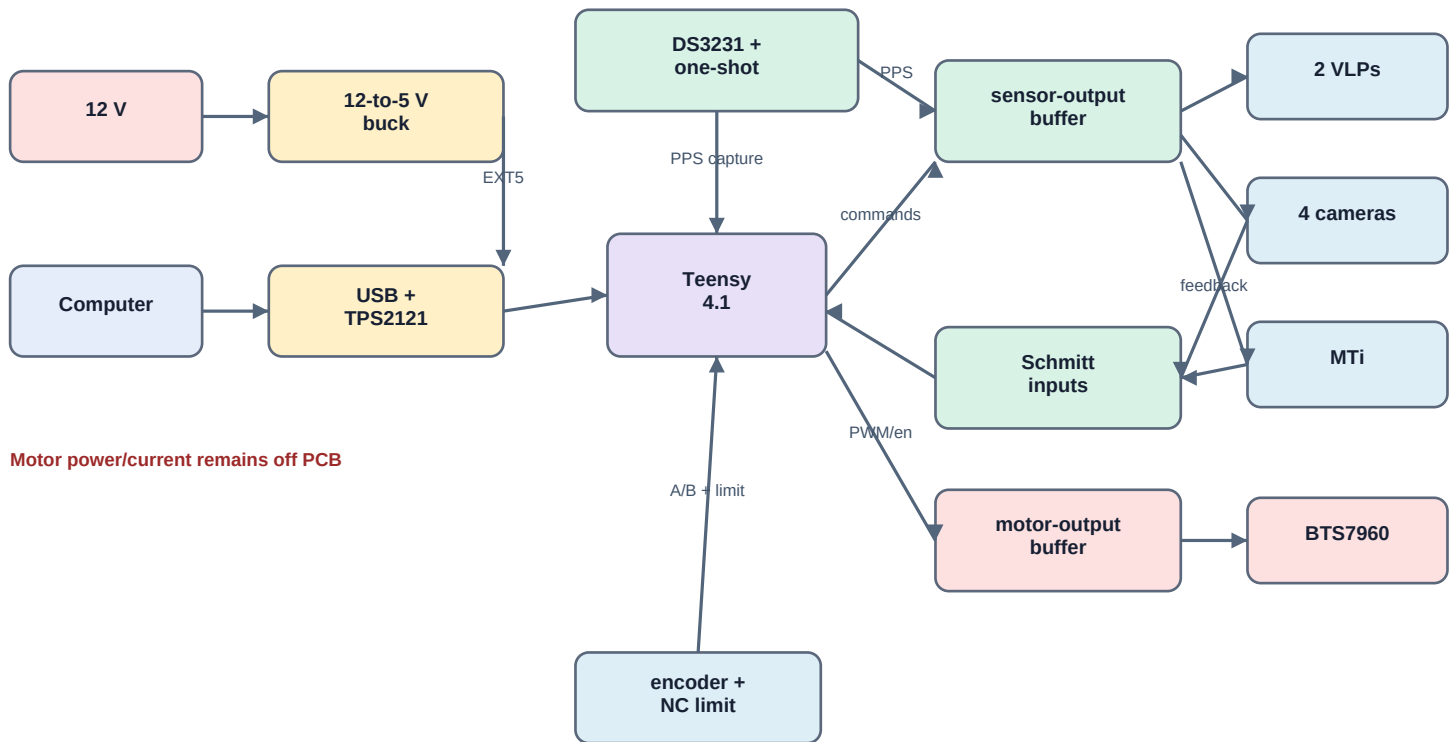
Requirements

V6 must:

- align two VLP-16s, four cameras, and one MTi to a defensible common local timebase;
- capture actual camera/IMU feedback events instead of using packet-arrival time;
- control the BTS7960 and count the existing 600 P/R A/B encoder;
- block motion toward the installed minimum when the normally-closed limit loop opens;
- default all motion outputs low during reset, faults, or missing commands;
- accept external 5 V as the preferred supply while USB backs up only the timing core;
- prevent 5 V backfeed and keep motor current out of PCB/USB/sensor grounds;
- power no PoE camera and carry no 12 V or motor power on this PCB;
- add no undefined knob, selector, or RC interface.

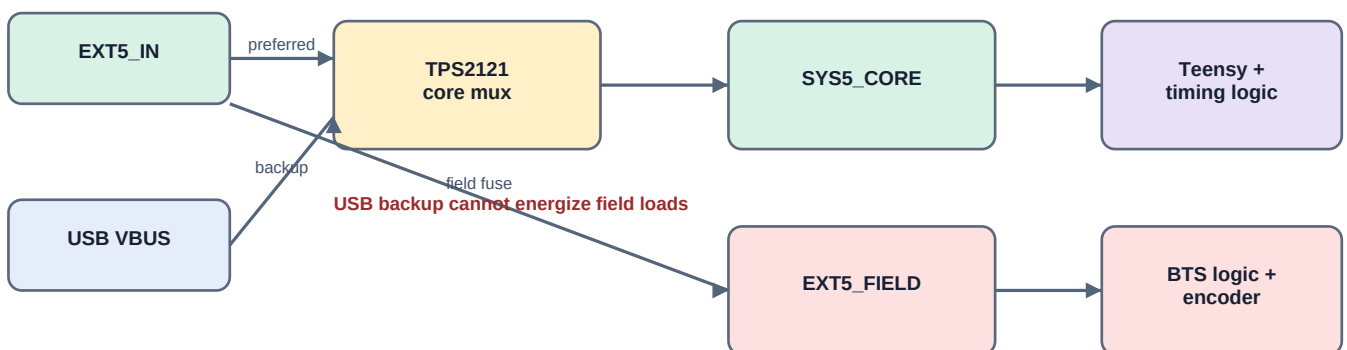
The DS3231 supplies a stable common local epoch, not GNSS-disciplined UTC.

Complete logical circuit



The PCB ground is a signal reference. BTS B-/B+/M+/M- use separate heavy conductors. Motor return current must never use a PCB trace, USB cable, sensor return, or Teensy ground pin.

Power, USB, and ground



- TPS2121 gives external-5-V priority, reverse-current blocking, current limiting, and USB backup for the core.
- USB-only operation must never energize EXT5_FIELD, encoder VCC, or BTS logic VCC.
- USB VBUS continues to the Teensy USB receptacle for USB attach behavior, but never joins VIN directly.
- Cut the Teensy 4.1 VUSB-VIN link and meter-prove the cut before connecting both sources.
- D27/A13 measures EXT5_IN; firmware forces motor enable low when field power is absent or out of range.

- Use 47-100 uF core bulk, 1 uF at the mux, and 100 nF at every IC. Final fuse values follow load and inrush measurements.

The anonymous external buck is accepted only after polarity, 4.75-5.25 V loaded output, ripple, temperature, maximum-load behavior, and input/output ground continuity are measured. This architecture is shared-ground, not isolated.

Teensy pin contract

Pin	Signal	Direction	Purpose
D1 / Serial1 TX	shared VLP NMEA	out	One hardware UART stream fans to both VLPs
D12	shaped PPS capture	in	Captures the same post-one-shot edge delivered to both VLPs
D11	common camera trigger	out	Four hardware-buffered branches
D14-D17	camera 1-4 feedback	in	Separate ExposureActive timestamps
D5 / D4	MTi SyncIn / SyncOut	out / in	IMU event command and evidence
D22 / D23	BTS RPWM / LPWM	out	PWM and direction under firmware control
D38	BTS common enable	out	Fans to R_EN and L_EN; default low
D30 / D31	encoder A / B	in	Quadrature capture
D34	minimum limit tripped	in	Open NC loop means trip/fault
D27 / A13	external 5 V monitor	in	Prevents USB backup from enabling motion
D18 / D19	RTC SDA / SCL	bidirectional	Native I2C at 3.3 V

Timed outputs and motion remain fail-closed until wiring, polarity, device settings, geometry, encoder sign, blocked motor direction, and hard E-stop operation are verified.

RTC and PPS one-shot

The owned Adafruit DS3231 product 3013 module and coin cell are retained. Its PCB envelope is approximately 22.86 x 17.78 mm with the exact header, holes, and battery clearance represented in the footprint. Power it from 3.3 V so its onboard I2C pull-ups cannot expose Teensy pins to 5 V.

The open-drain SQW output uses a 4.7 kOhm pull-up to 3.3 V and drives an SN74LVC1G123. A nominal 100 kOhm/100 nF network makes an approximately 10 ms PPS. That is pulse width, not start delay; the leading edge is delayed only by nanosecond-scale logic propagation. Scope-confirm the width and both loaded outputs.

Two VLP-16s

J2 pin	Function
1-3	VLP1 M12 pin 5 serial, pin 6 PPS, pin 8 signal return
4-6	VLP2 M12 pin 5 serial, pin 6 PPS, pin 8 signal return

The third conductor per VLP is required: do not use Ethernet shielding as the timing reference. One PPS fans to two protected branches; one UART NMEA stream fans to both LiDARs. Each branch uses connector-local TVS, a small series resistor, and a characterized load. Starting values are 180 Ohm series and 2.2 kOhm to ground.

Loaded outputs must satisfy VLP limits: high above 3.0 V and below 5.0 V, low below 1.2 V, and at least 2 mA high-state drive. Verify UART polarity on the exact Teensy core/VLP interface and never bit-bang NMEA. Emit nonblocking NMEA after the PPS trailing-edge margin and finish at least 300 ms before the next PPS.

Four Forge cameras

The PoE cameras receive no power from V6. Each camera uses four J5 conductors:

M8 pin	Function
2	OPTOIN configured as FrameStart
3	OPTOGND
6	OPTOOUT configured as ExposureActive
7	camera/output GND

The four-position pattern repeats for cameras 1-4, so J5 has 16 positions. Do not merge pins 3 and 7 without an authoritative circuit drawing and bench test.

Four output-driver channels create closely aligned trigger edges. Each branch has a default-low resistor, 100-220 Ohm series resistance, and connector-local TVS. Four separate 3.3 V, 5.5 V-tolerant Schmitt inputs capture OPTOOUT. Firmware records command and feedback; the camera's approximately 7-25 us optocoupler delay dominates the board's nanosecond-scale delay.

Xsens MTi

J3 pin	Function
1	Fischer pin 5 SyncIn
2	Fischer pin 6 SyncOut
3	common ground

SyncIn is a protected, buffered 5 V-compatible output. SyncOut feeds a protected, 3.3 V-powered, 5.5 V-tolerant Schmitt receiver. Firmware must account for any receiver inversion. The MTi remains continuously sampled; SyncIn marks a configured event and SyncOut provides evidence. Verify the exact installed MTi model and SyncSettings in MT Manager before enabling it.

BTS7960 motor interface

J101 pin	Function
1-4	RPWM, LPWM, R_EN, L_EN
5-6	EXT5_FIELD, GND

R_EN and L_EN share one buffered BTS_ENABLE. Connector-side pulldowns hold all commands low during reset or disconnection. Teensy firmware owns mutual exclusion, acceleration limiting, command timeout, break-before-make, limit direction gating, homing, and stall detection. The independent hard E-stop remains in the 12 V motor-power path. R_IS/L_IS are omitted until the exact IBT-2 clone has a verified current-sense contract.

Encoder

J102 pin	Function
1-4	EXT5_FIELD, GND, A/white, B/green

The Taiss encoder is a 5-24 V, 600 P/R, NPN open-collector A/B encoder; red is power, black ground, white A, green B, and there is no Z. A and B use 3.3 V pull-ups, TVS, small series resistors, and a dual Schmitt receiver. Teensy counts edges in interrupts and publishes bounded-rate state; it must not stream every edge. At 6300 rpm with x4 decoding, the worst case is about 252,000 edges/s.

Normally-closed limit

J103 pin	Function
1-2	NC-loop signal, GND/COM

The healthy closed loop reads inactive. Actuation, unplugging, or a broken wire opens it and reads tripped. Use pull-up, TVS, series resistance, 1-10 nF mechanical debounce, and a Schmitt receiver. This deliberate microsecond-scale filter is only on the mechanical safety input. Firmware blocks motion toward minimum while permitting controlled motion away.

Components and why they exist

Function	Preferred part	Reason
Core source selection	TPS2121	External priority, USB backup, reverse blocking, protection
PPS shaping	SN74LVC1G123	Converts 1 Hz SQW to repeatable short PPS
Sensor outputs	SN74AHCT244	3.3 V logic input and qualified 5 V-compatible sensor drive
Camera/MTi/limit inputs	SN74LVC14A	Six 5.5 V-tolerant Schmitt inputs
Encoder inputs	SN74LVC2G17	Two 5.5 V-tolerant non-inverting Schmitt inputs
Motor outputs	SN74AHCT125	Three default-low BTS commands plus spare
USB ESD	USBLC6-2SC6	Low-capacitance D+/D- protection
VLP/camera ESD arrays	TPD4E05U06	Four lines per connector-local package
MTi/encoder arrays	TPD2E2U06	Two lines per connector-local package
Limit ESD	TPD1E10B06	One protected line

Protection order is connector -> TVS with shortest ground return -> series resistor -> driver/receiver. A remote consolidated protection box would be ineffective. No RC filter is placed on PPS, NMEA, camera, MTi, or encoder timing. Board logic delay is single-digit to low tens of nanoseconds.

Measurement-time contract

Device	Command/reference	Returned evidence	Meaning
VLP x2	shaped PPS + common NMEA	sensor packet timestamp	common local RTC epoch, not GNSS UTC
Camera x4	common trigger	separate ExposureActive pulse	exposure event after characterized opto delay
MTi	SyncIn	SyncOut	configured IMU sync event
Encoder	none	counted A/B transitions	sampled position/velocity on Teensy monotonic time
Limit	none	accepted NC-loop transition	safety event after debounce

Records need a monotonic event ID, Teensy timestamp, channel, edge/state, scheduler status, and overflow/fault state. Host software maps the Teensy epoch to its own clock; it does not replace measurement time with USB/Ethernet arrival.

PCB placement

- J2 faces the adjacent LiDAR interface boards.
- J101/J102/J103 and the field fuse face the BTS/encoder/limit harness.
- J5 faces the camera bulkhead; J3 faces the MTi cable.
- USB, external 5 V, and TPS2121 share the service edge.
- Teensy is central; RTC, one-shot, and sensor-output buffer sit near D1/D11/D12.
- TVS arrays sit within millimeters of their connectors.
- Use four layers and a solid ground plane; route USB D+/D- first, then power, PPS/NMEA, camera/MTi, encoder, and motor/limit.
- Put tall parts on top and only small SMD parts underneath.
- Shrink the board only after full routing and cable-access review; do not force 100 x 64 mm at the expense of return paths.

No knob or switch is currently justified. Add labeled test points for all rails and major timing/motion nets. A future RC input may be a three-pin DNP reservation only after receiver voltage/protocol are known.

Required failure behavior

Failure	Behavior
USB lost, external 5 V healthy	core remains alive; command watchdog disables motor
External 5 V lost, USB remains	timing core may remain; field rail dies; motor enable forced low
Both sources absent	board off; hard E-stop remains independent
Teensy reset/high-Z	pulldowns hold RPWM, LPWM, enable low
Limit loop opens	tripped; motion toward minimum blocked
Encoder stops during motion	stall fault
Camera/MTi feedback missing	timing fault; no invented event
Event queue overflows	explicit latched fault

Bench and release gates

- [] Qualify the external buck and measure computer-ground to tray-star voltage before USB connection.
- [] Prove the Teensy VUSB-VIN cut.
- [] Test external-only, USB-only, and both-source cases with no backfeed; USB-only must not power field loads.
- [] Scope SQW, one-shot PPS, D12 capture, both loaded VLP PPS branches, and NMEA polarity/timing.
- [] Measure four-camera trigger skew and capture all ExposureActive returns.
- [] Verify the MTi model, settings, levels, and polarity.
- [] Test encoder maximum rate, sign, bounded telemetry, and stall behavior.
- [] Prove limit open-circuit, blocked-direction, reset, command-timeout, USB-loss, and hard-E-stop behavior.
- [] Replace the old isolated KiCad design; remove TMR0511, ISO7760s, FIELD_GND, isolation moat, and obsolete motor gates.
- [] Verify exact footprints, reach ERC = 0, route every net, and reach DRC = 0.

References

- TI TPS2121
- TI SN74LVC14A
- TI SN74LVC2G17
- TI SN74AHCT125
- Teledyne FLIR Forge FG-PGE-50S5-IP Technical Reference, Digital I/O Control
- Velodyne VLP-16 User Manual, GPS/PPS/NMEA electrical and timing requirements
- Xsens MTi User Manual and installed-model SyncSettings documentation

Maintenance rule

Markdown is the editable source of truth. After an electrical or firmware-interface change: update this file and revision/date; update KiCad and run ERC; update connector and Teensy contracts; update sensor_timing.md, telescope.md, and firmware configuration as needed; regenerate timing_circuit.pdf; visually inspect every page; and keep CAD/software evidence distinct from physical validation.